Certified coverage planning under execution uncertainty

1 Logged runs — target-field runs, or one run per other field

↓

2 Exact run score — S = largest gap or edge miss; zero miss ⇔ S ≤ λ

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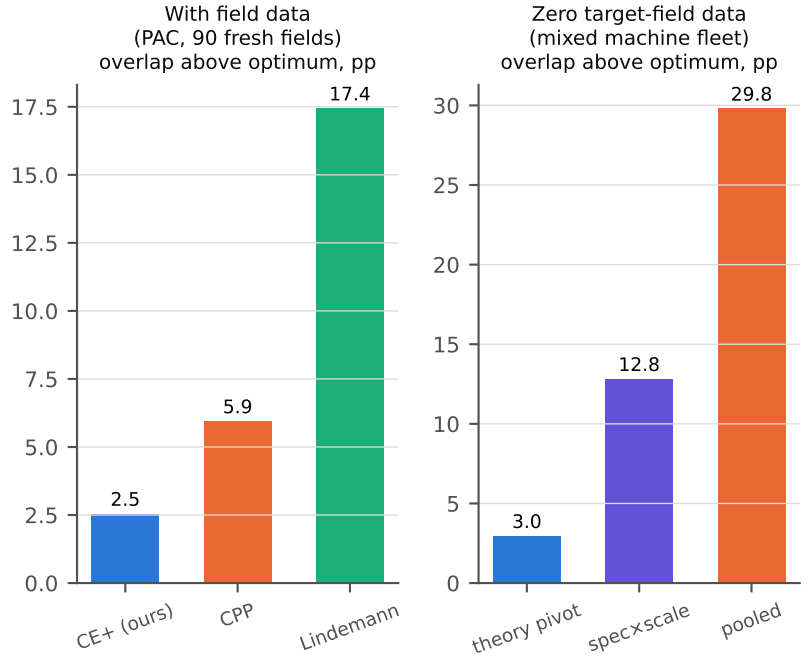
3 Five-layer theory — Lyapunov → Kac–Rice → product → mixture → Gumbel: each field's score quantile from controller, terrain, GNSS

↓

4 Conformal calibration — certified overlap λ: P(no miss) ≥ 1 − ε